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Computer Science and Engineering
ICS1513 – Computer Networks Laboratory

Experiment 4 - Inspecting HTTP packets using Wireshark Packet Sniffer

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I. The Basic HTTP Get/Response Interaction

1. Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

Both the browser and server are running HTTP/1.1 — the GET request reads GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1, and the response reads HTTP/1.1 200 OK.

2. What languages (if any) does your browser indicate that it can accept to the server?

The request shows Accept-Language: en-US,en;q=0.9 — so English (US) is preferred, with generic English as a fallback.

3. What is the IP address of your computer? Of the gaia.cs.umass.edu server?

End System (source): 10.116.29.233
Server (destination): 128.119.245.12

4. What is the status code returned from the server to your browser?

200 OK (visible in the packet list line for frame 325: HTTP/1.1 200 OK (text/html))

5. When was the HTML file that you are retrieving last modified at the server?

The file was last modified: Tue, 28 Oct 2025 05:59:01 GMT

6. How many bytes of content are being returned to your browser?

128 bytes (Content-Length: 128), matching the actual HTML content shown at the bottom of the packet.

7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.

Yes, headers like Sec-GPC: 1 or Upgrade-Insecure-Requests: 1 are visible directly in the raw hex/ASCII dump but not in Wireshark's parsed output.

8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

Yes — frame 412 (the GET for /wireshark-labs/HTTP-wireshark-file2.html) already includes:

If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT

If-None-Match: "173-64231b67176b7"

9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

No. The response (frame 414) is:

HTTP/1.1 304 Not Modified

10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes — frame 412 contains:

If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT

11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The response (frame 414) was:

HTTP/1.1 304 Not Modified

The status code is 304 Not Modified, which means the cached copy is still used. There's no Content-Length header and no body/File Data section in the response — only headers are visible

This confirms the browser used its cached copy of file2.html instead of the server re-sending the HTML.

12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

The browser sent one GET request for the Bill of Rights. It's in frame 19:

GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1

13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

Frame 25 contains the response headers, HTTP/1.1 200 OK, and the reassembled body.

14. What is the status code and phrase in the response?

200 OK

15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the the bill of rights.

4 segments

[4 Reassembled TCP Segments (4864 bytes): #22(1238), #23(1238), #24(1238), #25(1150)]

16. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

Your browser sent 4 GET requests total:

```
487  GET /wireshark-labs/HTTP-wireshark-file4.html  128.119.245.12 (gaia.cs.umass.edu)
496  GET /pearson.png    128.119.245.12 (gaia.cs.umass.edu)
500  GET /8E_cover_small.jpg    2.56.99.24 (kurose.cslash.net)
506  GET /favicon.ico      128.119.245.12 (gaia.cs.umass.edu)
```

Two embedded objects referenced in the HTML are pearson.png and 8E_cover_small.jpg — sent to two different servers,

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

In parallel. Here's the evidence from the timestamps:

Frame 496: GET /pearson.png sent at 14.853523635s

Frame 500: GET /8E_cover_small.jpg sent at 15.006384037s

Frame 504: response (301 redirect) for pearson.png arrives at 15.087315185s